LAUNDRY SERVICE BOOKING AND ORDER TRACKING SYSTEM

A PROJECT REPORT

Submitted by

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in partial fulfilment of the End Semester Examination for the course

CB23F35

LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF ENGINEERING
COMPUTER SCIENCE ENGINEERING



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BONAFIDE CERTIFICATE

Certified that this project report titled "**LAUNDRY SERVICE BOOKING AND ORDER TRACKING SYSTEM**" is the bonafide work of NITHISH RAJ L (240701366) who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

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DECLARATION

I, NITHISH RAJ L (240701366), hereby declare that the project report titled "**LAUNDRY SERVICE BOOKING AND ORDER TRACKING SYSTEM**", submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, dashboard, Deluge scripts, workflows, approval and blueprint — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature, code snippet or external resource used in this work has been duly acknowledged in the report.

SIGNATURE OF THE STUDENT

Place: Chennai
Date:

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NITHISH RAJ L

ABSTRACT

Small service businesses such as laundries still depend on notebooks, phone calls and memory to record customers, calculate charges and follow the progress of each order. These methods are slow and error-prone: the price of an order is worked out by hand, the status of a bundle of clothes is known only to the person who received it, and daily or monthly totals have to be compiled manually. The Laundry Service Booking and Order Tracking System replaces this with a single, structured application built on Zoho Creator, the low-code platform used in the course CB23F35.

The application contains two linked forms, *Customer details* and *Laundry Orders*. A laundry order records the customer (through a lookup), service type, weight, price per kg, order, pickup and delivery dates, order status, discounted amount and final amount. Twelve form workflows written with Deluge calculate the total amount from weight and price, apply a 10% discount to orders of ₹500 or more, validate weight, price and dates before saving, and generate an order confirmation message. A blueprint moves every order through the stages Received, Washing, Ready and Delivered, and an approval workflow lets the CEO-level approver accept or reject an order, with an email sent to the submitter. Four list reports (all customers, all orders, pending orders and completed orders) and a dashboard with order counts, revenue, an Orders by Status chart and a Revenue by Service Type chart give the owner an immediate view of the business.

The application was tested with ten functional test cases covering customer registration, mandatory-field and weight/date validation, amount calculation, discount, blueprint status change, reports and dashboard; all ten passed. It was built and tested as a single-user prototype on a Zoho Creator trial account, with the owner acting under the CEO role. Role-based access for staff, delivery agents and customers, payment tracking and customer feedback are proposed as future enhancements. The project shows that a complete order-management workflow can be built on a low-code platform with only small amounts of Deluge scripting.

Keywords: Low-Code Development; Zoho Creator; Laundry Service; Order Tracking; Deluge Scripting; Blueprint; Workflow Automation.

PROJECT AT A GLANCE

Attribute	Details
Project Title	Laundry Service Booking and Order Tracking System
Application Domain	Service Industry / Small-Business Order Management (Laundry)
Student Name & Register No.	NITHISH RAJ L – 240701366
Zoho Creator Edition	Trial account
Application Name (Link Name)	laundry-service-booking-and-order-tracking-system
Published / Live URL	Not published; accessed through the Zoho Creator account
Demo Credentials (per role)	Not applicable (single-user prototype; owner login)
Number of Forms / Subforms	2 / 0 (Customer details; Laundry Orders)
Number of Reports (by type)	4 list reports (All Customer Details, All Laundry Orders, Pending Laundry Orders, Completed Laundry Orders)
Number of Pages / Dashboards	1 dashboard (Laundry Service Dashboard) with 4 KPI cards and 2 charts
Number of Workflows	17 configured: 12 form workflows, 3 schedules (currently disabled), 1 approval, 1 blueprint
Deluge Functions / Custom Actions	12 form-workflow actions; 5 scripts documented in full (Appendix C); no stand-alone custom functions
Integrations	None external; email notification through the approval ‘Send mail’ action
Roles & Profiles Defined	Single user; application owner under the CEO role (approver). Staff, Delivery Agent and Customer roles are proposed
Environments Used	Single (trial) environment
Date of Production Deployment	Not applicable (academic prototype)

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CEO	Chief Executive Officer (role used by the application owner)
CRUD	Create, Read, Update, Delete
ER	Entity–Relationship
FR / NFR	Functional / Non-Functional Requirement
KPI	Key Performance Indicator
LCAP	Low-Code Application Platform
RBAC	Role-Based Access Control
SDLC	Software Development Life Cycle
UAT	User Acceptance Testing
UI	User Interface
Zia	Zoho’s AI assistant

CHAPTER 1
INTRODUCTION

1.1 Background

The project belongs to the service-industry domain, specifically the day-to-day operation of a small laundry. A typical laundry receives clothes from walk-in customers, washes, irons or dry-cleans them, and returns them on an agreed date. The owner must remember which customer handed over which bundle, how much it weighs, what the charge is, when it must be delivered and whether it is ready. In many small shops this information lives in a notebook or in the memory of the staff. This project digitises that process in a Zoho Creator application so that customer details, orders, charges, dates and status are kept in one place.

1.2 Problem Statement

Manual laundry record-keeping makes it hard to know what work is pending, what a customer owes and what the business has earned. Charges are calculated by hand each time, discounts are applied inconsistently, and customers have to call to find out whether their clothes are ready. The Laundry Service Booking and Order Tracking System addresses these problems by providing a centralised, searchable record of customers and orders, automatic amount calculation, and a clear order status that moves through fixed stages.

- Customer and order details are written on paper and are easily lost or mis-read.
- Total charges and discounts are calculated manually and may be inconsistent.
- There is no quick way to see pending orders, delivered orders or total revenue.
- Order progress is communicated verbally, so the status of a bundle is not visible to everyone.

1.3 Objectives

- To design a structured data model with forms for customers and laundry orders linked by a lookup field.
- To use Deluge form workflows to calculate the total amount, apply a discount and validate weight, price and dates.
- To track every order through the stages Received, Washing, Ready and Delivered using a blueprint, with an approval step and an order confirmation message.
- To provide list reports and a dashboard showing pending orders, completed orders, revenue and order distribution.
- To test the application with realistic sample data and document its security and governance settings.

1.4 Scope of the Project

In scope are customer registration, laundry order booking, automatic total, discount and final-amount calculation, input validation, order status tracking through a blueprint, an approval workflow, an order confirmation message, list reports and a dashboard. The application is built as a single-user prototype in which the owner works under the CEO role. Out of scope for this version are online payment collection, a separate payment and feedback module, delivery-agent and customer logins, SMS or WhatsApp notifications and inventory management. These are described as future enhancements in Chapter 9.

1.5 Justification for a Low-Code Approach

Zoho Creator is appropriate because the application is mainly built from forms, relationships between records, reports and simple business rules. Its visual form builder, report builder and dashboard builder allow a working prototype to be created quickly, while Deluge handles the calculations and validations that configuration alone cannot express. The platform provides hosting, a database and user authentication, so the student can concentrate on the data model and workflows instead of building a web server separately [1].

1.6 Organisation of the Report

Chapter 2 reviews the existing system and the literature. Chapter 3 presents the requirement analysis. Chapter 4 describes the system design. Chapter 5 details the implementation on Zoho Creator. Chapter 6 covers testing and debugging. Chapter 7 discusses deployment, performance, security and governance. Chapter 8 presents the results and discussion, and Chapter 9 concludes the report with future enhancements.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In a typical small laundry the customer hands over clothes at the counter and the staff writes the customer's name and the items in a register. The charge is worked out by hand from the weight and the rate for the chosen service. When the customer asks about the order, the staff check the register or ask the person doing the washing. At the end of the day or month the owner adds up the entries to find the sales and the number of pending orders. There is no single record that links a customer to an order, its charge, its dates and its current status.

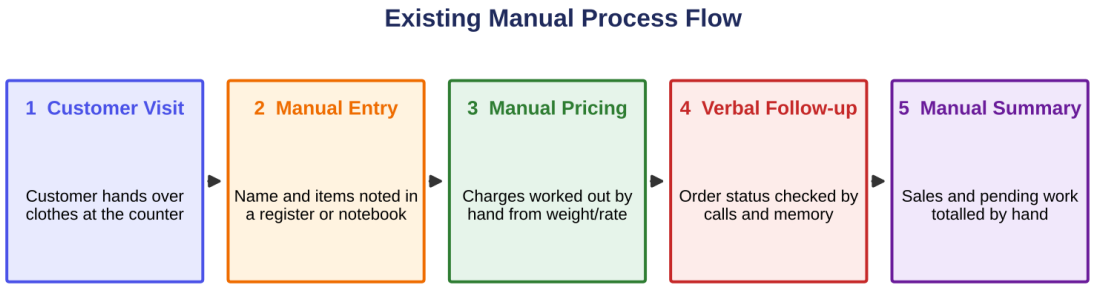


Figure 2.1: Existing Process Flow — Customer Visit → Manual Entry → Manual Pricing → Verbal Follow-up → Manual Summary

2.2 Limitations of the Existing System

- Records are on paper and can be lost, duplicated or become outdated.
- Charges and discounts are calculated manually, which can lead to mistakes.
- There is no real-time view of pending and delivered orders.
- Revenue and workload summaries must be compiled by hand.

2.3 Low-Code versus Traditional Development

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development time	Requires coding the UI, database, authentication and hosting	Visual forms, reports and dashboards shorten prototype development
Skills required	Programming, database, web framework and deployment skills	Form design, data modelling and basic Deluge/workflow knowledge
Hosting & infrastructure	Developer selects and maintains hosting and database	Hosting and database provided by the platform, subject to edition
Mobile application	Needs responsive design or separate mobile development	Desktop, tablet and mobile views are available from the platform

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Security & access control	Must be designed and coded	Built-in login, users, roles and permission settings
Maintenance & change	Code changes and redeployment	Many form and report changes can be made visually
Customisation limits	High flexibility with enough effort	Bound by platform features, Deluge capabilities and plan limits
Cost	Development and hosting costs vary	Subscription or trial limits apply

2.4 Literature Survey

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	Zoho Corporation, n.d.	Zoho Creator Help Documentation [2]–[5]	Documents forms, reports, workflows, approvals and application configuration.	Primary reference for building the tracker in Zoho Creator.
2	Zoho Corporation, n.d.	Deluge Help [6]	Explains Deluge syntax and the functions supported for application logic.	Supports the calculation, validation and message scripts.
3	Sahay, A., Indamutsa, A., Di Ruscio, D., Pierantonio, A., 2020	Supporting the Understanding and Comparison of Low-Code Development Platforms [7]	Studies low-code development platforms and proposes a way to understand and compare them by their characteristics, such as visual development, model-driven techniques and supported features.	Provides the research background for choosing and describing Zoho Creator as a low-code platform.

2.5 Emerging Trends in Low-Code Development

Low-code development increasingly combines visual application building with configurable automation, reusable components, analytics and AI-assisted creation. For this project, visual forms and reports form the core of the application, while Deluge is used only where conditional logic is needed, such as the discount rule and the validations. A dashboard presents counts and revenue. These trends led to keeping the first version small and modular, leaving advanced analytics, notifications to customers and payment integration as future enhancements.

2.6 Proposed System

The proposed system is a Zoho Creator application that centralises laundry order management. The owner registers a customer, then books a laundry order by choosing the customer and service type and entering the weight, price per kg and the order, pickup and delivery dates. Workflows calculate the total amount, apply the discount and the final amount, and block invalid entries. A blueprint moves the order from Received to Washing, Ready and Delivered, an approval workflow lets the CEO accept or reject an order, and reports and a dashboard show pending and completed orders and revenue. All data is stored in one database, which improves traceability and removes repeated manual calculation.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application	Status
Owner / Administrator (CEO role)	Application owner with full configuration access	Register customers, book orders, update status through the blueprint, approve orders, view reports and dashboard	Implemented (single user)
Staff	Counter or washing staff	Create bookings and update order progress	Proposed
Delivery Agent	Person who delivers finished orders	View assigned deliveries and update delivery status	Proposed
Customer	Person who hands over laundry	Provides laundry and receives the order confirmation; may later view own orders through a portal	External / proposed

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall allow a customer to be registered with name, phone, email and address; name, phone and address are mandatory.	Owner	High
FR-02	The system shall allow a laundry order to be booked for a registered customer with service type, weight, price per kg and order, pickup and delivery dates.	Owner	High
FR-03	The system shall calculate the total amount automatically from weight and price per kg.	Owner	High
FR-04	The system shall apply a 10% discount when the total amount is ₹500 or more and calculate the final amount.	Owner	High
FR-05	The system shall reject a weight of zero or less and a delivery date earlier than the pickup date.	Owner	High
FR-06	The system shall track each order through the stages Received, Washing, Ready and Delivered.	Owner	High
FR-07	The system shall provide an approval step for orders and send an email to the submitter on approval or rejection.	Owner (CEO)	Medium
FR-08	The system shall provide reports of all customers, all orders, pending orders and completed orders.	Owner	Medium
FR-09	The system shall provide a dashboard showing total, pending and completed orders, revenue, orders by status and revenue by service type.	Owner	Medium
FR-10	The system shall generate an order confirmation message when an order is saved.	Owner	Medium

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	For the small expected dataset, forms, reports and the dashboard should open and update without noticeable delay.
NFR-02	Security	Only a signed-in Zoho user with access to the application can view or change records.
NFR-03	Usability	Forms should use clear labels, required-field indicators and drop-down lists for service type and status.
NFR-04	Availability	Availability depends on the Zoho Creator service, the account and internet connectivity.
NFR-05	Maintainability	Forms, fields and workflows use descriptive names so that logic can be understood and changed easily.
NFR-06	Accuracy	Amounts must be calculated by the application and not typed manually.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator (trial account)
Scripting language	Deluge
Browser	Google Chrome / Microsoft Edge / Mozilla Firefox, current supported version
Other Zoho services	None connected; email is sent through the built-in approval action
Mobile	Creator mobile/tablet layout views (device testing not performed)
Client hardware	Computer or mobile device with a supported browser and internet connection

3.5 Use Case Model

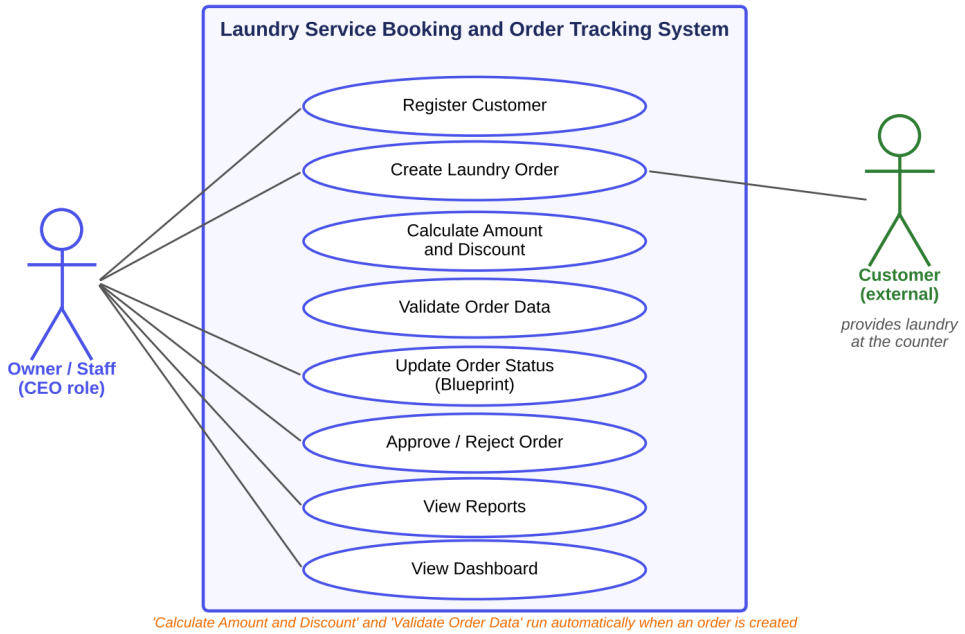


Figure 3.1: Use Case Diagram of the Laundry Service Booking and Order Tracking System

Table 3.5: Use Case Description – Book a Laundry Order

Use Case ID	UC-01
Use Case Name	Book a Laundry Order
Actor(s)	Owner / Staff (CEO role)
Pre-condition	The customer is registered in Customer details and the owner is signed in.
Main Flow	1. Open Laundry Orders. 2. Select the customer and service type. 3. Enter weight and price per kg; the total amount, discount and final amount are filled automatically. 4. Enter order, pickup and delivery dates and the order status. 5. Submit. 6. The order is saved, a confirmation message is generated and the order appears in the reports.
Alternate Flow	If the weight is zero or less, or the delivery date is earlier than the pickup date, an ‘Invalid entries found’ message is shown and the order is not saved.
Post-condition	The order is stored with its amounts and status and can be moved through the blueprint stages.
Related FR	FR-02, FR-03, FR-04, FR-05, FR-10

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

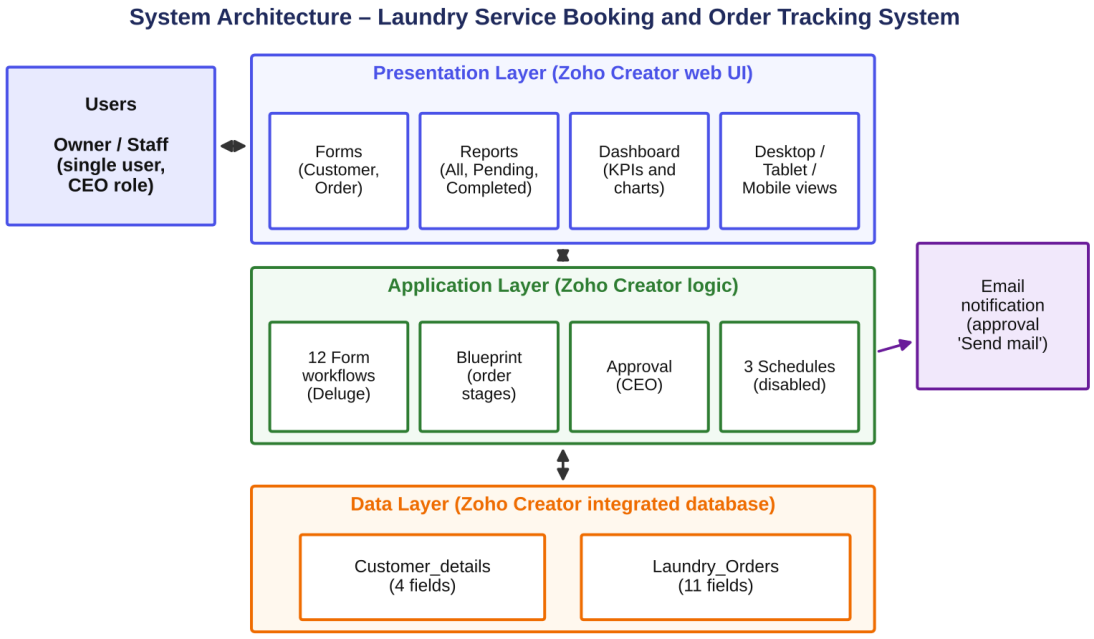


Figure 4.1: System Architecture of the Laundry Service Booking and Order Tracking System

The system has three logical layers. The presentation layer is the Zoho Creator web interface, which provides forms, reports and the dashboard and can be viewed in desktop, tablet and mobile layouts. The application layer contains the form workflows written in Deluge, the blueprint, the approval workflow and the schedules. The data layer is the Zoho Creator integrated database, which stores the Customer_details and Laundry_Orders forms with a lookup relationship. An email notification is sent through the approval workflow's 'Send mail' action.

4.2 Data Model

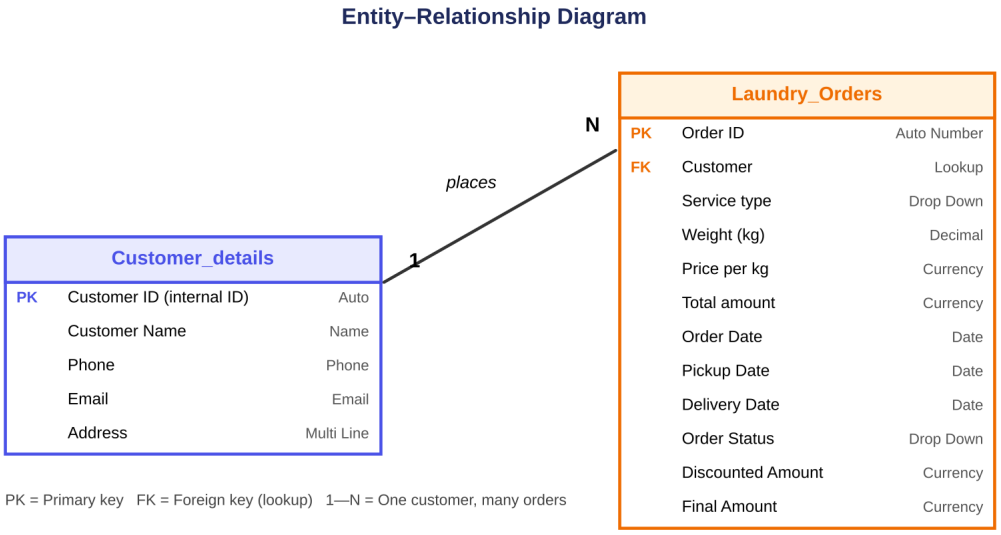


Figure 4.2: Entity–Relationship Diagram

Table 4.1: Entities and Relationships

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
Customer_details	Customer Name, Phone, Email, Address	Laundry_Orders	One customer has many orders (one-to-many)	Lookup field in Laundry_Orders
Laundry_Orders	Order ID, Customer, Service type, Weight (kg), Price per kg, Total amount, dates, Order Status, Discounted Amount, Final Amount	Customer_details	Many orders belong to one customer	Lookup field ‘Customer’

4.3 Form Design

Table 4.2: Field Specification – Customer details Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Customer Name	Customer_Name	Name	Yes	Required	Customer’s first name (shown as first_name in scripts)
Phone	Phone	Phone	Yes	Required; country code +91 default	Contact number
Email	Email	Email	No	Email format	Optional contact
Address	Address	Multi Line	Yes	Required	Customer address

Table 4.3: Field Specification – Laundry Orders Form

Field Label	Link Name	Field Type	Validation / Default	Remarks
Order ID	Order_ID1	Auto Number	System generated	Unique order number
Customer	Customer	Lookup	Selects from Customer details	Links order to customer
Service type	Service_type	Drop Down	Washing, Ironing, Wash and Iron, Dry Cleaning	Type of service
Weight (kg)	Weight_kg	Decimal	Must be greater than 0 (Validate Laundry Weight)	Laundry weight
Price per kg	Price_per_kg	Currency	Validated (Validate Price per kg)	Rate for the service
Total amount	Total_amount	Currency	Calculated = weight × price per kg	Before discount
Order Date	Order_Date	Date	Date picker	Date of booking
Pickup Date	Pickup_Date	Date	Checked against delivery date	Collection date
Delivery Date	Delivery_Date	Date	Not earlier than pickup date	Promised delivery
Order Status	Order_Status	Drop Down	Received, Ready, Delivered (and Washing through the blueprint)	Current stage
Discounted Amount	Discounted_Amount	Currency	10% of total if total ≥ ₹500, else 0	Discount value
Final Amount	Final_Amount	Currency	Total amount − discount	Amount payable

4.4 Process and Workflow Design

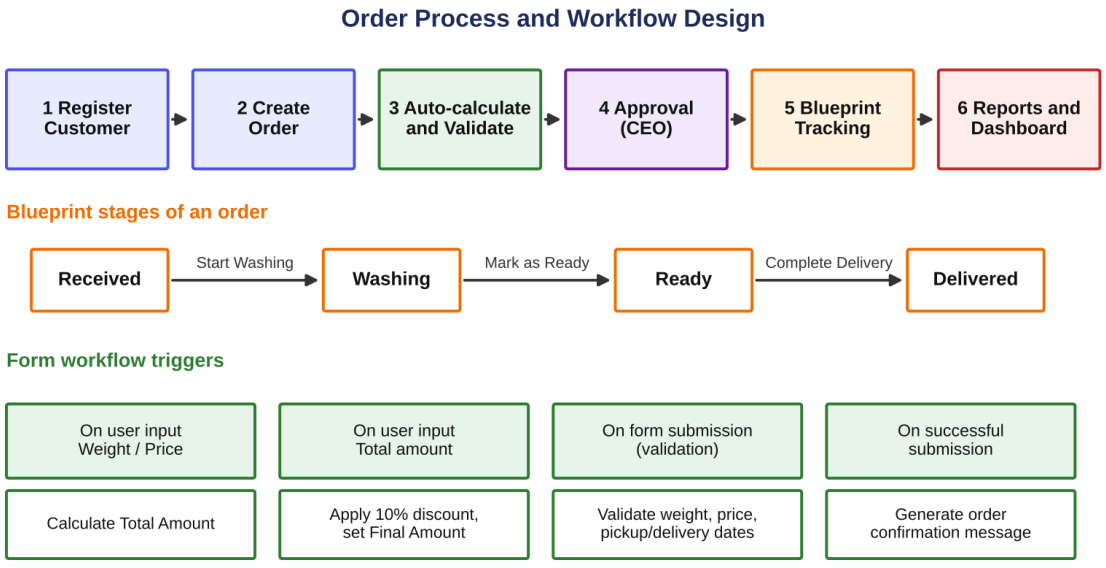


Figure 4.3: Order Process and Workflow Design

The overall flow is: register the customer, create the order, let the workflows calculate and validate, send the order for approval where required, track it through the blueprint stages and view the result in the reports and dashboard. The blueprint allows only the transitions Start

Washing (Received to Washing), Mark as Ready (Washing to Ready) and Complete Delivery (Ready to Delivered), so an order cannot skip a stage.

4.5 User Interface and Navigation Design

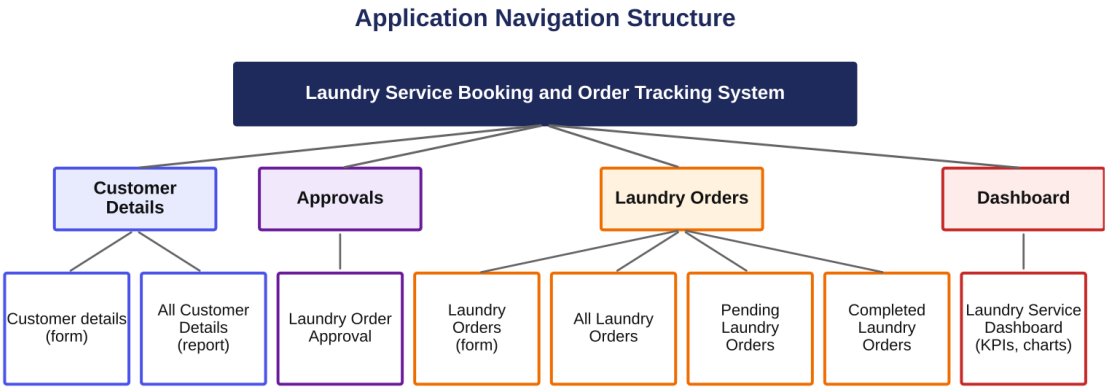


Figure 4.4: Navigation Structure of the Application

The left menu groups the application into Customer Details, Approvals, Laundry Orders and the Laundry Service Dashboard. Each group contains the form and the reports that belong to it, so the owner can move between data entry, reports and the dashboard with one click.

4.6 Access Control Design

The prototype is a single-user application: the application owner signs in under the CEO role and has full access to all forms, reports and the dashboard, and is also the Level 1 approver of the approval workflow. The matrix below shows the implemented access and the access proposed for additional roles in a future multi-user version.

Table 4.4: Role–Permission Matrix (A = Add, E = Edit, V = View, D = Delete)

Component	Owner (CEO) – implemented	Staff – proposed	Delivery Agent – proposed	Customer – proposed
Form: Customer details	A E V D	A E V	—	V (own)
Form: Laundry Orders	A E V D	A E V	V (assigned); update delivery status	V (own)
Reports (all / pending / completed)	V	V	V (assigned)	V (own)
Dashboard	V	V	—	—
Approval: Laundry Order Approval	Approve / Reject	—	—	—

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The application “Laundry Service Booking and Order Tracking System” (link name laundry-service-booking-and-order-tracking-system) was created in Zoho Creator on a trial account using the native form and report builders; Deluge scripts, a blueprint and an approval were then added. The left menu is organised into Customer Details, Approvals, Laundry Orders and the Laundry Service Dashboard, as shown in Figure 5.1.

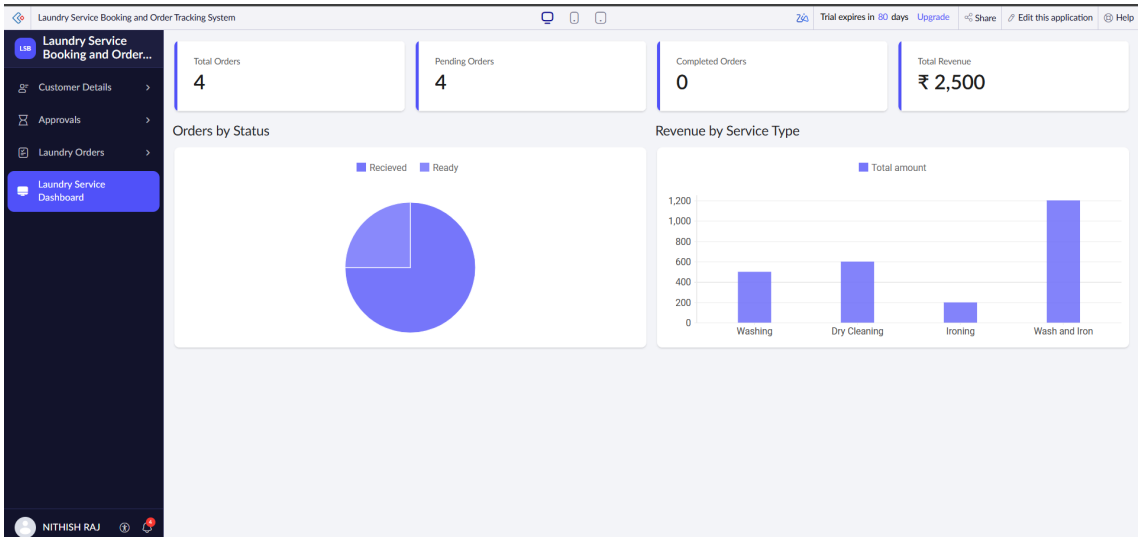


Figure 5.1: Laundry Service Booking and Order Tracking System – Home and Navigation

5.2 Forms and Data Capture

The **Customer details** form stores the customer name, phone, email and address; name, phone and address are mandatory. The **Laundry Orders** form stores the order. Its fields are an Auto Number order ID, a Customer lookup, a Service type drop-down (Washing, Ironing, Wash and Iron, Dry Cleaning), Weight (kg) as a decimal, Price per kg, Total amount, Discounted Amount and Final Amount as currency fields, the Order, Pickup and Delivery dates, and an Order Status drop-down. Together the two forms use ten distinct field types. The form builder views are shown in Figures 5.2 and 5.4, and the live forms in Figures 5.3 and 5.5.

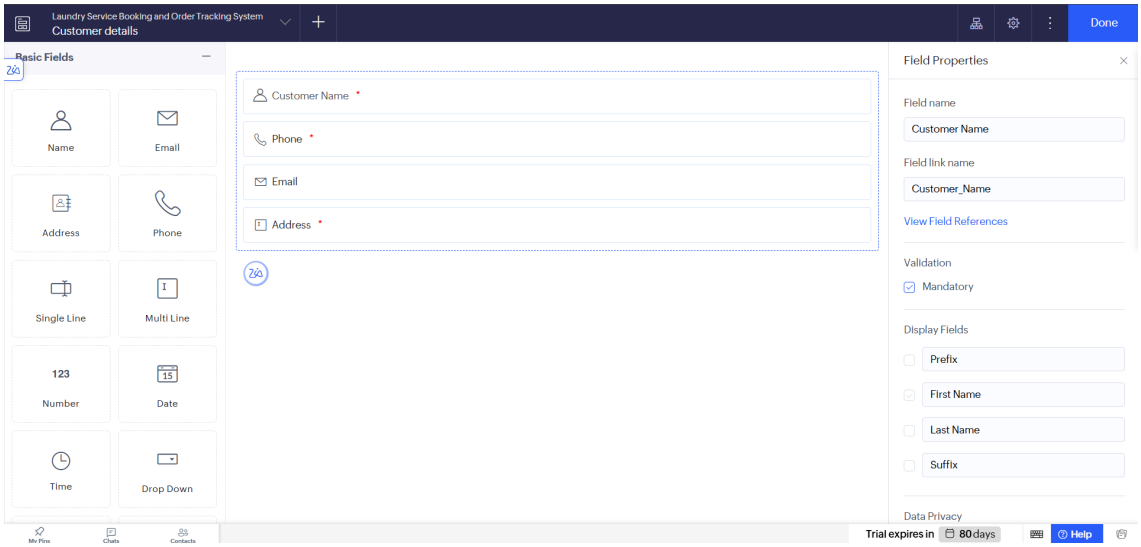


Figure 5.2: Customer details Form in the Form Builder

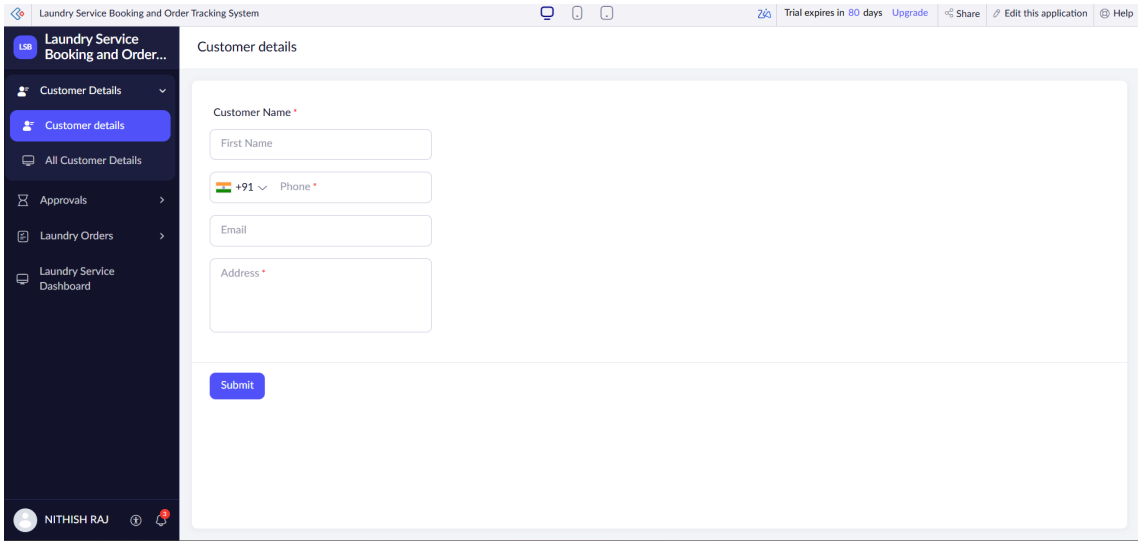


Figure 5.3: Customer details Form in Live Mode

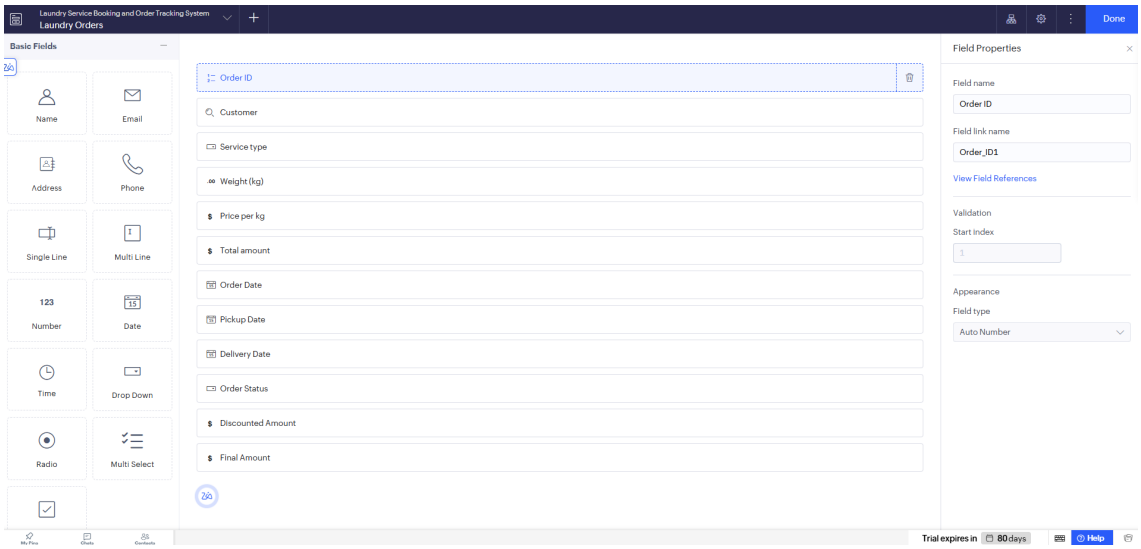


Figure 5.4: Laundry Orders Form in the Form Builder

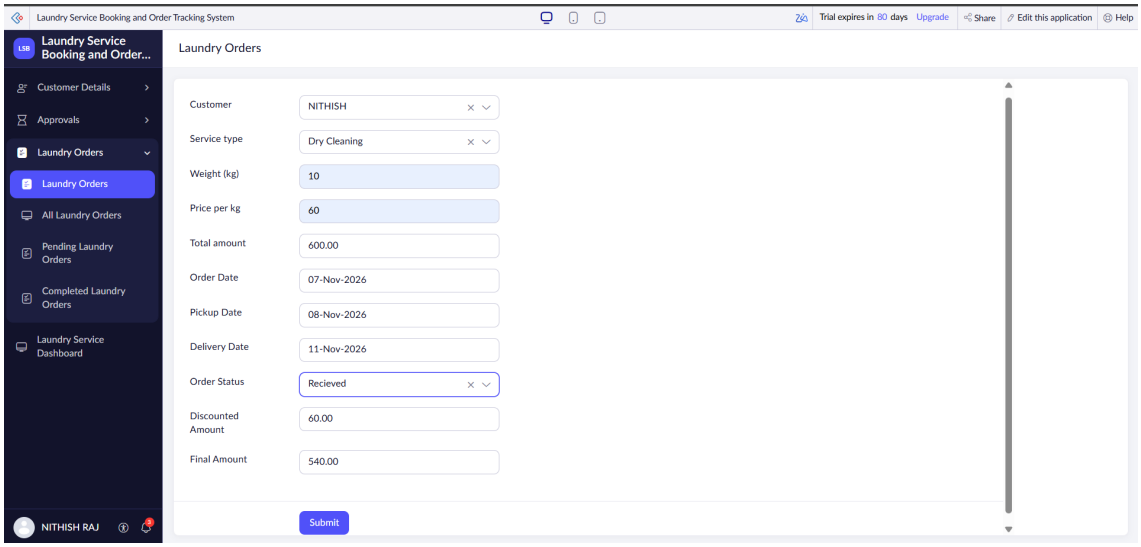


Figure 5.5: Laundry Orders Form Filled with Sample Data (10 kg × ₹60 = ₹600; discount ₹60; final amount ₹540)

5.3 Reports

Table 5.1: Reports Implemented

Report Name	Type	Source Form	Filter / Grouping / Sorting	Custom Actions	Roles
All Customer Details	List	Customer_details	All records	Open, add, edit	Owner
All Laundry Orders	List	Laundry_Orders	All orders; columns Order ID, Customer, Service type, Order Status, Weight, Price per kg, Total amount	Open, add, edit	Owner
Pending Laundry Orders	List	Laundry_Orders	Orders not yet delivered (Received / Ready)	Open, add, edit	Owner
Completed Laundry Orders	List	Laundry_Orders	Orders with status Delivered	Open, add, edit	Owner
Orders by Status	Pie chart (dashboard)	Laundry_Orders	Grouped by Order Status	—	Owner
Revenue by Service Type	Bar chart (dashboard)	Laundry_Orders	Total amount grouped by Service type	—	Owner

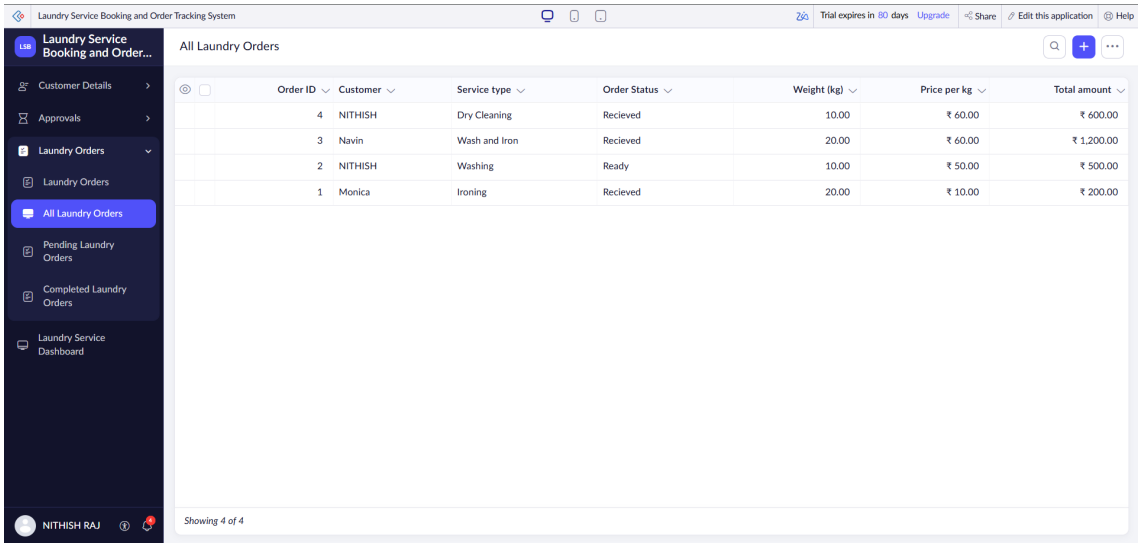


Figure 5.6: All Laundry Orders Report

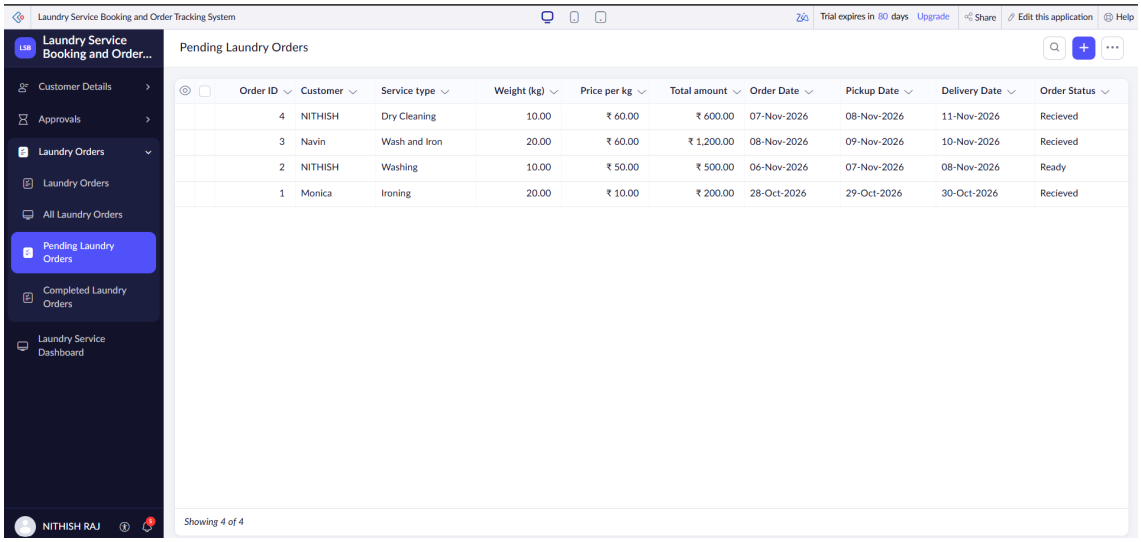


Figure 5.7: Pending Laundry Orders Report

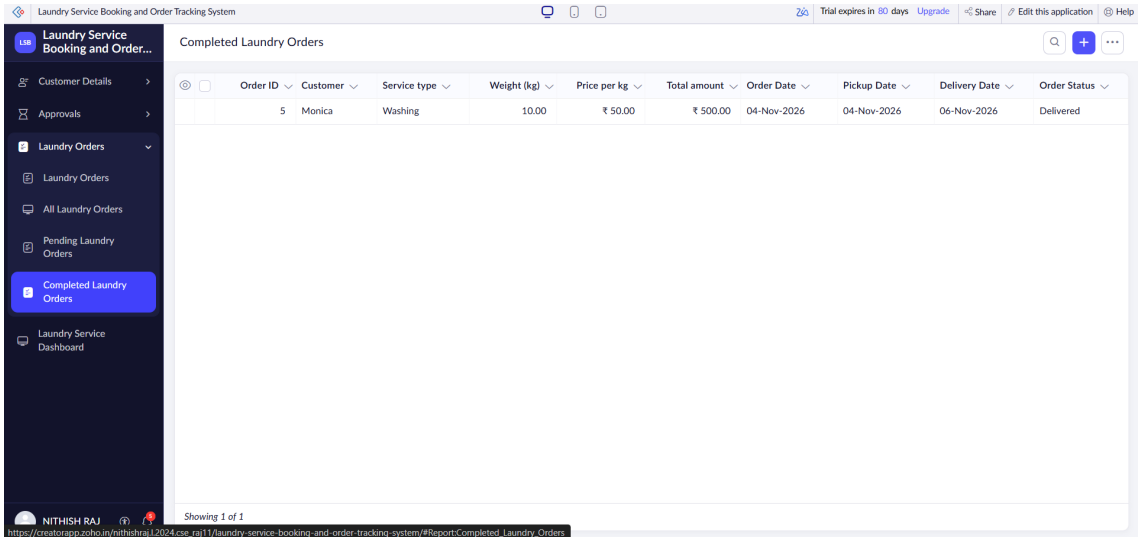


Figure 5.8: Completed Laundry Orders Report

5.4 Pages and Dashboards

The Laundry Service Dashboard gives the owner a one-screen summary. Four KPI cards show Total Orders, Pending Orders, Completed Orders and Total Revenue. Revenue is the sum of the Total amount field, that is, before the discount. The ‘Orders by Status’ pie chart shows how orders are distributed across statuses and the ‘Revenue by Service Type’ bar chart compares Washing, Dry Cleaning, Ironing and Wash and Iron. The figure below was captured when four orders had been saved (four pending, none completed, ₹2,500 total amount: 200 + 500 + 1,200 + 600); the dashboard reads the saved records each time it is opened. The charts are based on saved data only and do not imply any measure of service quality.

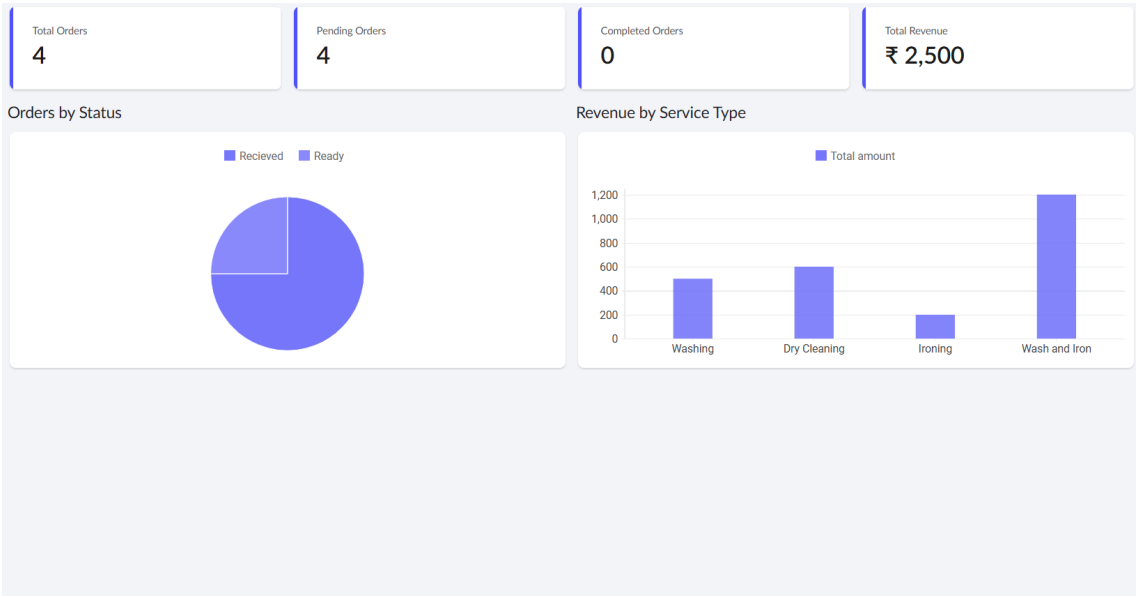


Figure 5.9: Laundry Service Dashboard

5.5 Deluge Scripting

Deluge is used in the form workflows of the Laundry Orders form. Five scripts are documented in full in Appendix C and their editor views are shown in Figures 5.10 to 5.14.

Table 5.2: Deluge Scripts Implemented

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-01	Laundry Orders – On user input of Weight (kg): Calculate Total Amount on Weight Input	Multiplies weight by price per kg and stores the result in Total amount; sets 0 if either value is missing	input fields, isNull(), if/else, arithmetic
DS-02	On user input of Total amount: Apply Laundry Discount	Gives a 10% discount when the total is ₹500 or more, sets Discounted Amount and calculates Final Amount	if/else, arithmetic, field assignment
DS-03	On form submission: Validate Laundry Weight	Rejects a weight of zero or less	if, alert, cancel submit
DS-04	On form submission: Validate Pickup and Delivery Dates	Rejects a delivery date earlier than the pickup date	nested if, date comparison, alert, cancel submit

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-05	On successful form submission: Generate Order Confirmation Message	Fetches the customer record by ID and builds a multi-line confirmation message with order details	record fetch with criteria, string concatenation

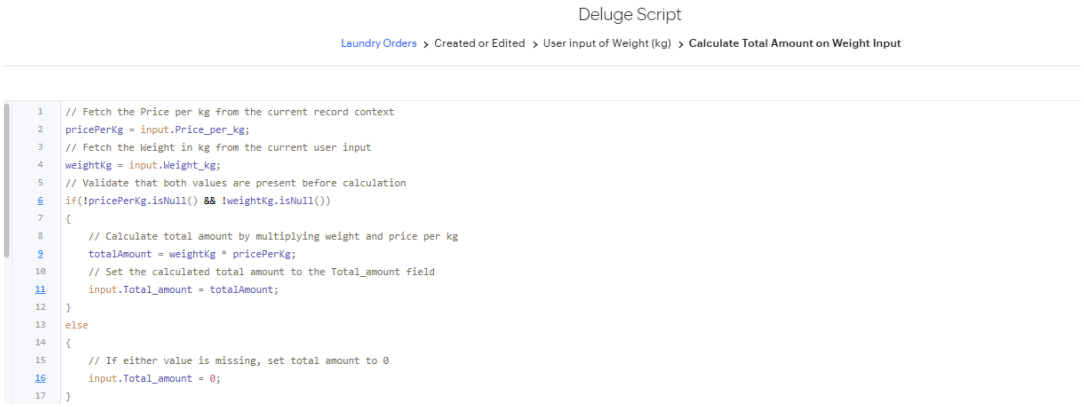


Figure 5.10: DS-01 – Calculate Total Amount on Weight Input

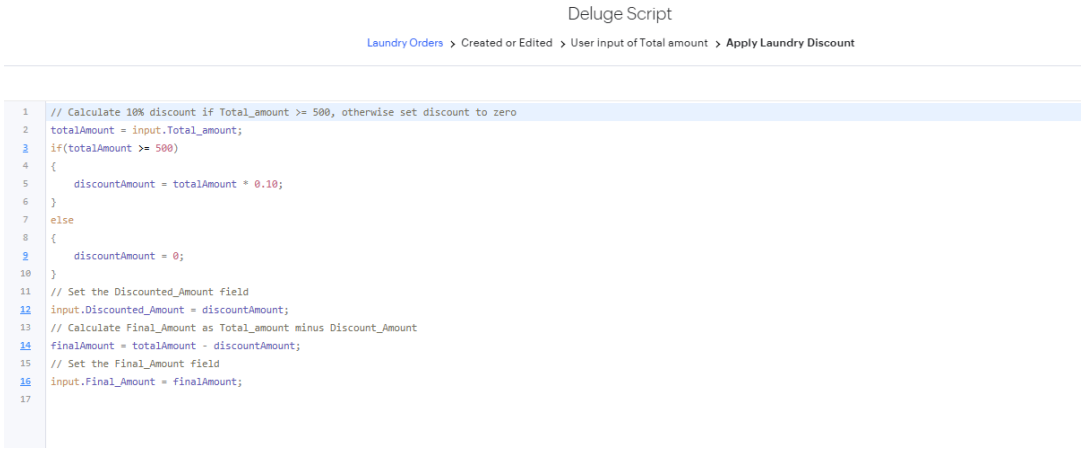


Figure 5.11: DS-02 – Apply Laundry Discount



Figure 5.12: DS-03 – Validate Laundry Weight



Figure 5.13: DS-04 – Validate Pickup and Delivery Dates

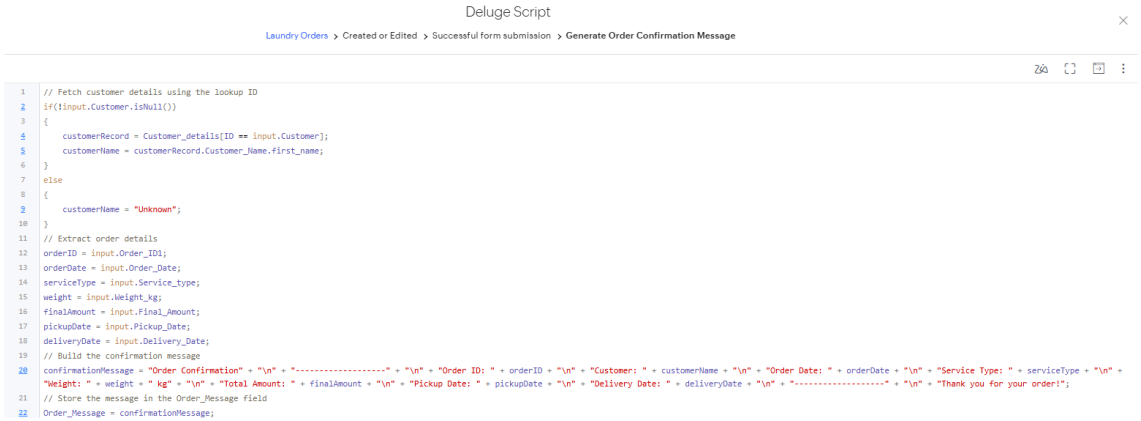


Figure 5.14: DS-05 – Generate Order Confirmation Message

The total-amount script is triggered when the weight is entered. Because the discount script is triggered by a change in Total amount, the discount and final amount update automatically after every recalculation, which is how an order of 10 kg at ₹60 per kg shows ₹600, ₹60 and ₹540 in Figure 5.5. The two validation scripts use *alert* to display a message against the field and *cancel submit* to stop the record from being saved.

5.6 Workflows and Automation

The application contains 17 configured workflows: 12 form workflows, 3 schedules, 1 approval and 1 blueprint (Figures 5.15 to 5.20). Table 5.3 lists the form workflows with their triggers as shown in the workflow editor. The scripts of DS-01 to DS-05 are documented in Appendix C; for the other workflows the purpose is given from the workflow name.

Table 5.3: Form Workflows Implemented (Laundry Orders form)

No.	Workflow Name	Trigger / Event	Purpose	Script Ref.
1	Flag Overdue Laundry Orders	Edited – update of Delivery Date	Flags orders whose delivery date has passed	—
2	Flag Overdue Laundry Orders	Edited – update of Order Status	Re-checks the overdue flag when status changes	—
3	Identify High-Value Orders	Edited – update of Total amount	Identifies orders with a high total amount	—
4	Calculate Delivery Days on Delivery Date Input	Created or edited – user input of Delivery Date	Calculates delivery days when the delivery date is entered	—
5	Calculate Delivery Days on Pickup Date Input	Created or edited – user input of Pickup Date	Calculates delivery days when the pickup date is entered	—
6	Calculate Total Amount on Price Input	Created or edited – user input of Price per kg	Recalculates the total when the price changes	—
7	Apply Laundry Discount	Created or edited – user input of Total amount	10% discount for totals of ₹500 or more	DS-02
8	Calculate Total Amount on Weight Input	Created or edited – user input of Weight (kg)	Total = weight × price per kg	DS-01
9	Validate Laundry Weight	Created or edited – validation on form submission	Weight must be greater than zero	DS-03
10	Validate Price per kg	Created or edited – validation on form submission	Checks the price per kg before saving	—
11	Validate Pickup and Delivery Dates	Created or edited – validation on form submission	Delivery date not earlier than pickup date	DS-04
12	Generate Order Confirmation Message	Created or edited – successful form submission	Builds the order confirmation message	DS-05

Table 5.4: Schedules, Approval and Blueprint Implemented

Workflow Name	Type	Trigger / Event	Condition / Action	Status
Daily Laundry Remainder	Schedule	Custom schedule, 09:30 (first run 31-Oct-2026)	1 action	Configured, currently disabled
Weekly Laundry Report	Schedule	Custom schedule, 09:30 (first run 31-Oct-2026)	1 action	Configured, currently disabled
Monthly Laundry Summary	Schedule	Custom schedule, 09:30 (first run 01-Nov-2026)	1 action	Configured, currently disabled
Laundry Order Approval	Approval	On review of a Laundry Orders record	Level 1 approver: CEO. On Approve and on Reject: send mail to the submitter (2 actions)	Enabled
Laundry Order Process	Blueprint	Status transition	Received → Start Washing → Washing → Mark as Ready → Ready → Complete Delivery → Delivered	Enabled

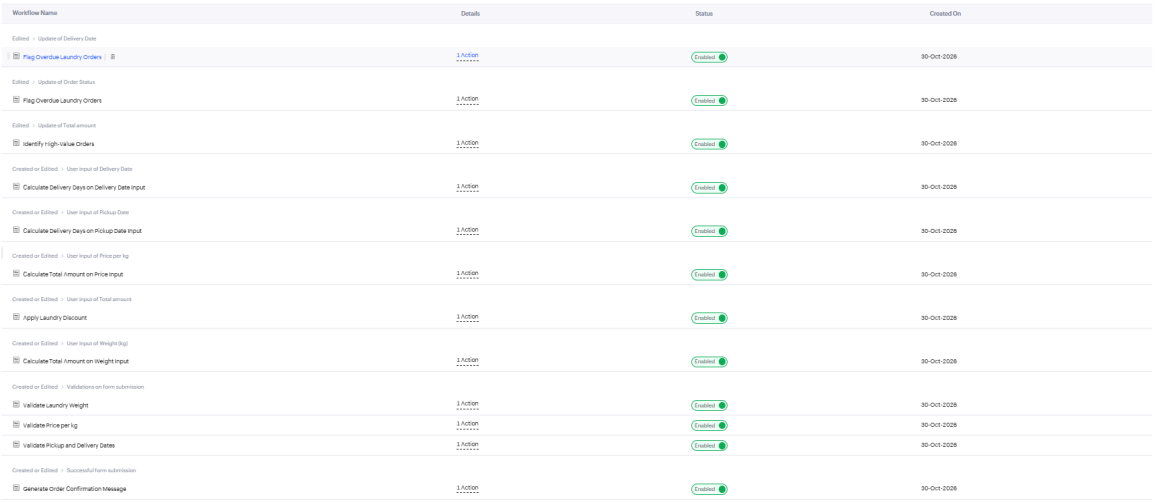


Figure 5.15: Form Workflows of the Laundry Orders Form

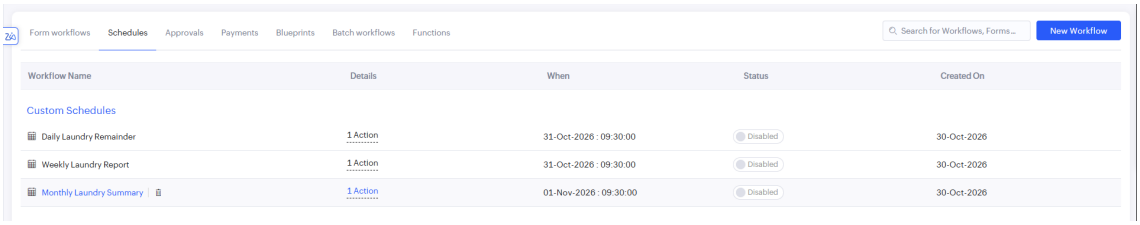


Figure 5.16: Schedules

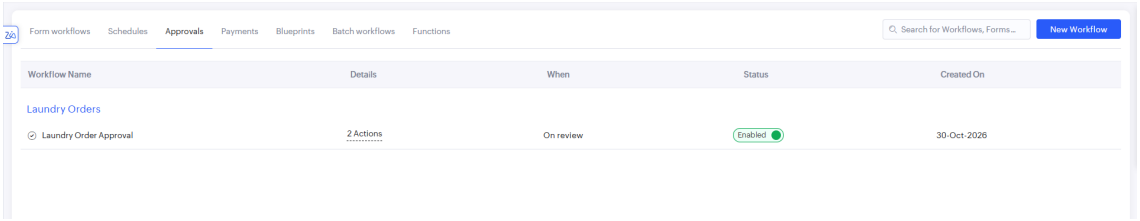


Figure 5.17: Approvals List

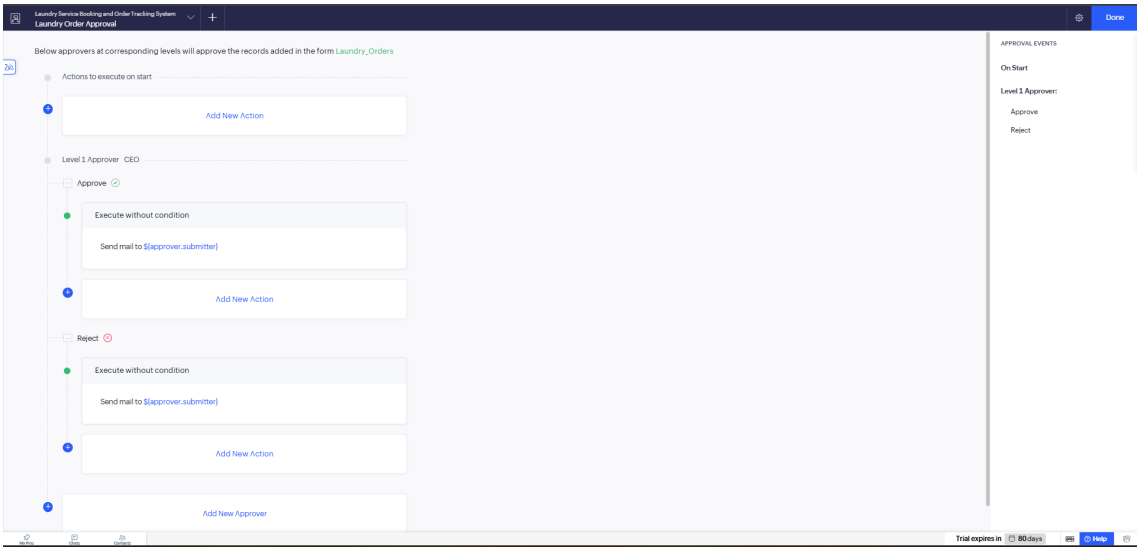


Figure 5.18: Laundry Order Approval Configuration (Level 1 Approver: CEO)

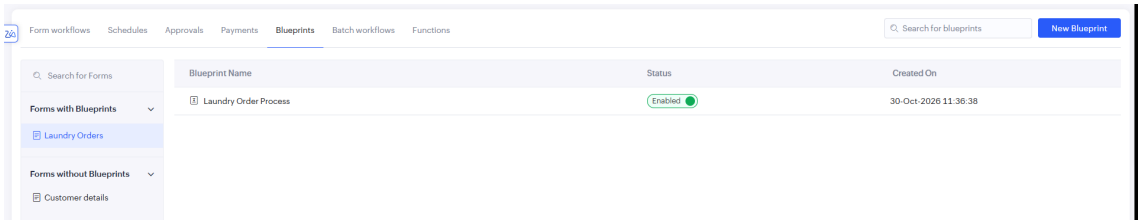


Figure 5.19: Blueprints List

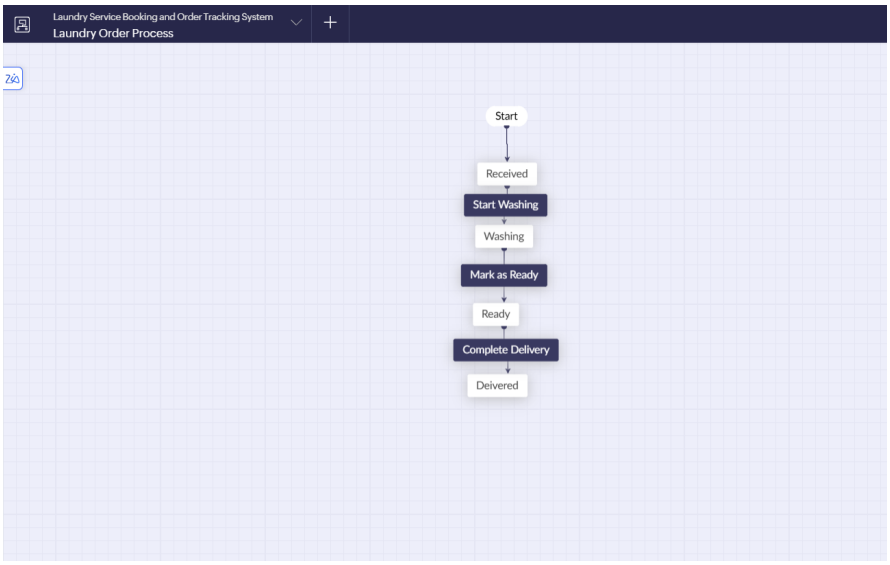


Figure 5.20: Laundry Order Process Blueprint

During use, a record that is in a blueprint stage shows only the transition button allowed from that stage, for example the ‘Start Washing’ button on a record in the Received stage (Figure 6.1). This guarantees that every order follows the same sequence.

5.7 Integrations

Table 5.5: Integrations Implemented

Integration	Service / Endpoint	Method	Authentication	Data Exchanged / Purpose
Email notification	Built-in ‘Send mail’ action of the approval workflow	Workflow action	Creator account	Sends mail to the submitter when an order is approved or rejected
External API	None	—	—	No external API is used in this version
Webhook / Creator REST API	None	—	—	Possible future extension for payment or messaging services

5.8 User Management and Role-Based Access

The application was built and tested as a single-user prototype. The application owner signs in with a Zoho account and holds the CEO role at the top of the role hierarchy; the same role is the Level 1 approver of the Laundry Order Approval workflow. Because only one user exists, record-level permissions and separate profiles were not required. The roles Staff, Delivery Agent and Customer, with the permissions in Table 4.4, are part of the design and are proposed for a multi-user version.

5.9 AI and Analytics Features

Analytics are provided by the native dashboard: four KPI cards and two charts (Orders by Status and Revenue by Service Type). No Zia or other AI feature is used in the running application; AI-assisted analysis is listed as a future enhancement.

5.10 Pro-Code Extension

No pro-code extension is included in this version. Zoho Creator forms, reports, the dashboard and Deluge are sufficient for the order-tracking workflow. A future version could use the Creator REST API, a custom widget or Zoho Catalyst for payment gateway integration or advanced analytics.

5.11 Mobile and Portal Access

The Zoho Creator application can be viewed in desktop, tablet and mobile layouts using the view icons at the top of the application (visible in Figures 5.1 and 5.9). No customer portal has been configured in this version; a portal for customers to place and track orders is a future enhancement.

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

Testing covered the forms, the lookup relationship, field validation, calculations, the blueprint, the reports and the dashboard. Positive tests confirmed that valid customers and orders are saved and appear in reports. Negative tests confirmed that missing mandatory values, a weight of zero and a delivery date earlier than the pickup date are rejected. Each test was run in Zoho Creator using sample customers (Monica, NITHISH and Navin) and sample orders.

6.2 Test Cases

Table 6.1: Test Cases

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-01	FR-01	Register a customer with valid details	Enter name, phone, email and address; submit	Customer is saved	Customer saved and available in the Customer lookup	Pass
TC-02	FR-01	Submit customer without mandatory fields	Leave name, phone or address blank; submit	Mandatory-field validation appears	Record not saved until required fields were filled	Pass
TC-03	FR-02	Book an order with valid values	Select customer, service, weight 10, price 60, dates; submit	Order is saved	Order 4 saved and listed in All Laundry Orders	Pass
TC-04	FR-05	Weight of zero	Enter weight 0; submit	Submission blocked with a message	‘Invalid entries found’ shown; order not saved (Fig. 6.2)	Pass
TC-05	FR-05	Delivery date earlier than pickup date	Pickup 09-Nov-2026, delivery 08-Nov-2026; submit	Submission blocked	‘Invalid entries found’ shown; order not saved (Fig. 6.3)	Pass
TC-06	FR-03	Total amount calculation	Weight 10 kg, price ₹60	Total = ₹600	Total amount shown as 600.00	Pass
TC-07	FR-04	Discount rule	Total ₹600 (≥ ₹500)	Discount ₹60; final ₹540	Discounted Amount 60.00 and Final Amount 540.00	Pass
TC-08	FR-06	Blueprint status change	Open order in Received stage	‘Start Washing’ offered as next step	Button shown on the record (Fig. 6.1)	Pass

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-09	FR-08	Pending and completed reports	Open Pending and Completed Laundry Orders	Undelivered orders in Pending; Delivered in Completed	Pending lists Received/Ready orders; Completed lists the Delivered order	Pass
TC-10	FR-09	Dashboard figures	Open Laundry Service Dashboard	KPIs and charts match saved orders	4 orders, 4 pending, 0 completed, ₹2,500; charts match the list	Pass

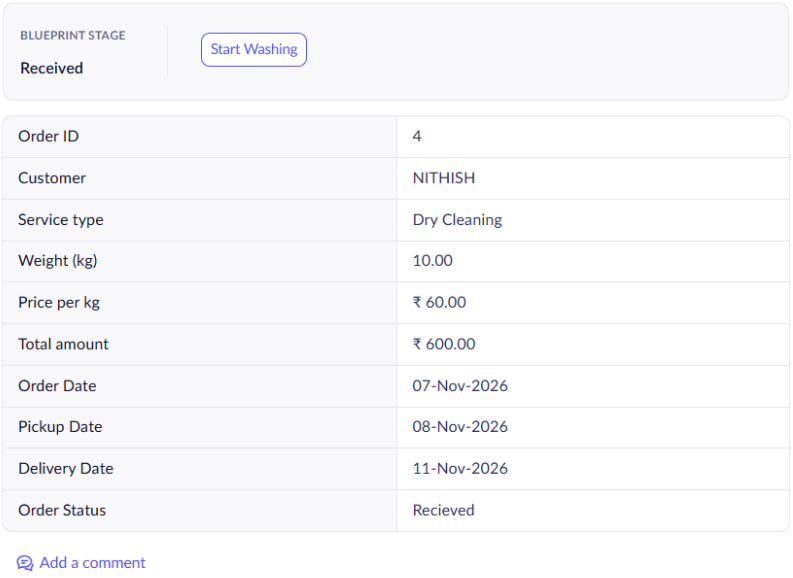


Figure 6.1: Blueprint Stage and ‘Start Washing’ Button on an Order Record

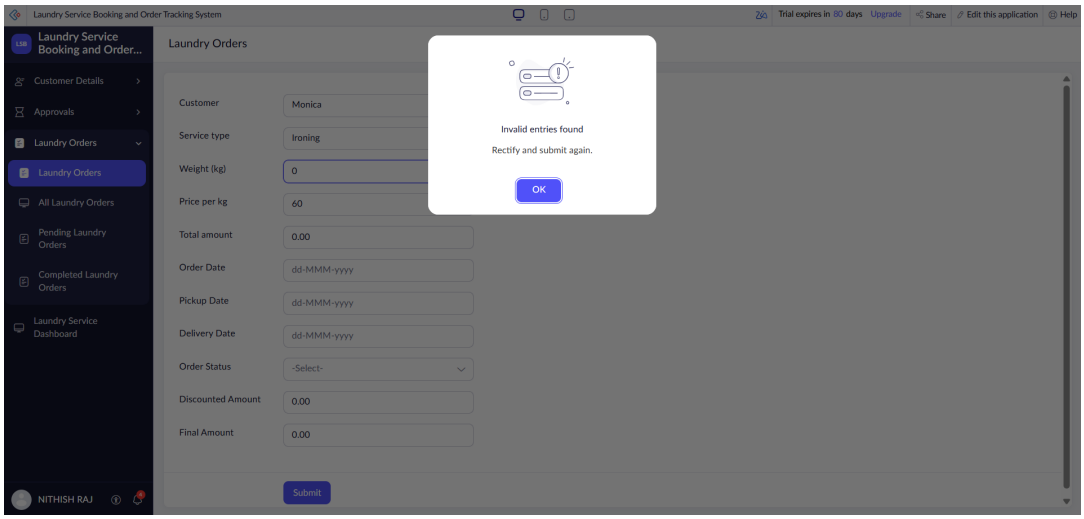


Figure 6.2: Validation Message for a Weight of Zero (TC-04)

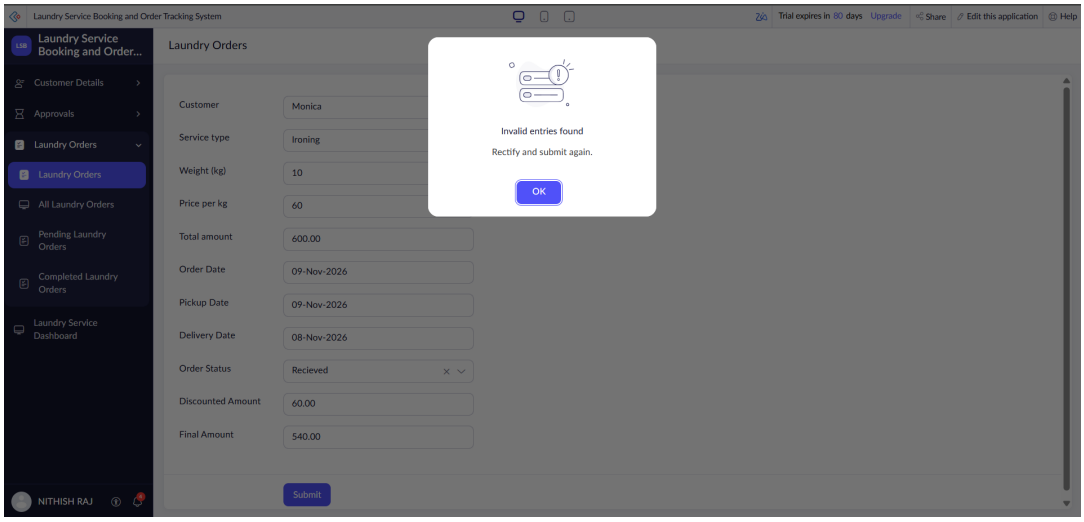


Figure 6.3: Validation Message for a Delivery Date Earlier than the Pickup Date (TC-05)

6.3 Debugging Deluge Scripts

Scripts were built and tested one at a time. The Deluge editor's syntax checking was used while writing each script, and each workflow was then tested with a small set of records by entering valid and invalid values in the live form. Because several workflows act on the same form (weight, price and total amount), they were tested in sequence: first the total amount, then the discount that depends on it, then the validations, and finally the confirmation message on a successful save. Field link names such as Weight_kg, Price_per_kg and Order_ID1 were checked in the form builder so that the scripts refer to the correct fields (Figure 5.4).

6.4 Defect Log

Table 6.2: Defect Log

Bug ID	Description	Root Cause	Fix Applied	Status
—	No functional defect was observed in the executed test cases. The 'Invalid entries found' messages in TC-04 and TC-05 are the intended behaviour of the validation workflows.	—	—	Closed

6.5 User Acceptance Testing

Table 6.3: User Acceptance Feedback

User	Role Tested	Feedback	Action Taken
Developer / application owner	Owner (CEO role)	Booking is faster because the total, discount and final amount are filled automatically; pending and completed reports and the dashboard reflect saved orders; the blueprint makes the order stage clear.	None required

User	Role Tested	Feedback	Action Taken
Further users	Staff / Customer	Not applicable in the single-user prototype; to be collected when the role-based version is deployed.	Planned

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

The application was developed and tested in a single Zoho Creator environment on a trial account and accessed through the 'Access this application' option. Separate development, staging and production environments were not used, so no multi-stage deployment pipeline is claimed. For a real laundry the application would be moved to a paid edition, the sample data removed, and the owner and staff users added.

Table 7.1: Deployment Checklist

S.No.	Checklist Item	Status
1	All test cases executed (TC-01 to TC-10)	Done
2	Sample/test data (5 sample orders, 3 sample customers) retained for demonstration	Done; to be removed before real use
3	Roles, permissions and users configured	Single owner user under the CEO role
4	Connections authorised	Not applicable (no external integration)
5	Schedules and workflows enabled	12 form workflows, approval and blueprint enabled; 3 schedules configured but disabled
6	Application accessible through the Creator account	Done
7	Version / release notes recorded	Not applicable for the prototype

7.2 Performance Optimisation

The dataset is small, so no timing measurements were taken and no numerical improvement is claimed. The following design choices keep the application light:

Table 7.2: Performance Considerations

Optimisation Applied	Component	Effect
Customer name fetched by a single record criteria (ID == input.Customer)	DS-05 Generate Order Confirmation Message	Only one record is read; no loop over all customers
Calculations done on the field the user is editing	DS-01 and DS-02	Totals update immediately without a separate report or script run
Reports filtered by status	Pending / Completed Laundry Orders	Each report loads only the records it needs
Schedules kept disabled until needed	Daily, Weekly and Monthly schedules	No unnecessary background executions in the trial account

7.3 Security Features

Table 7.3: Security Controls Implemented

Security Control	Where Applied	Threat Mitigated
Authenticated access	Zoho account login required to open the application	Unauthorised access to customer and order data
Mandatory fields	Customer name, phone and address	Incomplete customer records
Input validation (Deluge)	Weight, price per kg, pickup and delivery dates	Invalid or inconsistent order data
Approval workflow	Laundry Orders, CEO as Level 1 approver	Unreviewed changes to orders
Role hierarchy	CEO role at the top of the hierarchy	Defines the owner's authority; further roles to be added

7.4 Governance and Administration

Forms, fields and workflows have descriptive names (for example Calculate Total Amount on Weight Input and Validate Pickup and Delivery Dates) so that the logic can be understood and maintained. The application owner is responsible for reviewing the workflows, enabling the schedules when they are needed, removing test data and adding users when the application is used by more than one person. Where the edition permits, application data should be exported periodically as a backup.

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

Demonstration scenario: (1) the owner registers a customer; (2) a laundry order is booked, and the total, discount and final amounts are filled in automatically; (3) the order is moved through the blueprint stages and appears in the Pending or Completed report and in the dashboard. Figures 8.1 to 8.3 show these three stages. Table 8.1 lists the sample orders saved in the application.

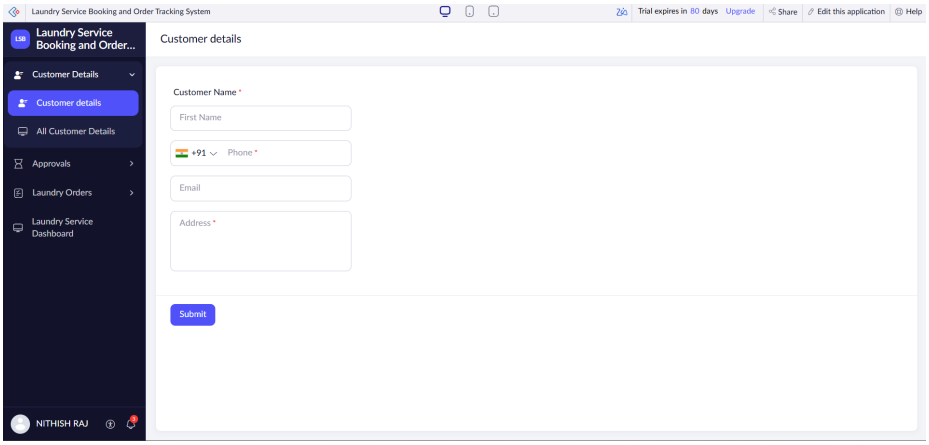


Figure 8.1: Step 1 — Owner Registers a Customer

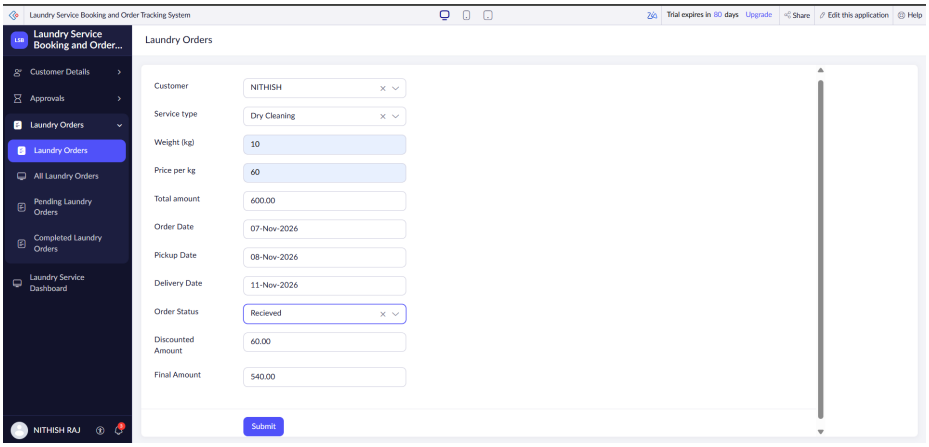


Figure 8.2: Step 2 — Owner Books a Laundry Order

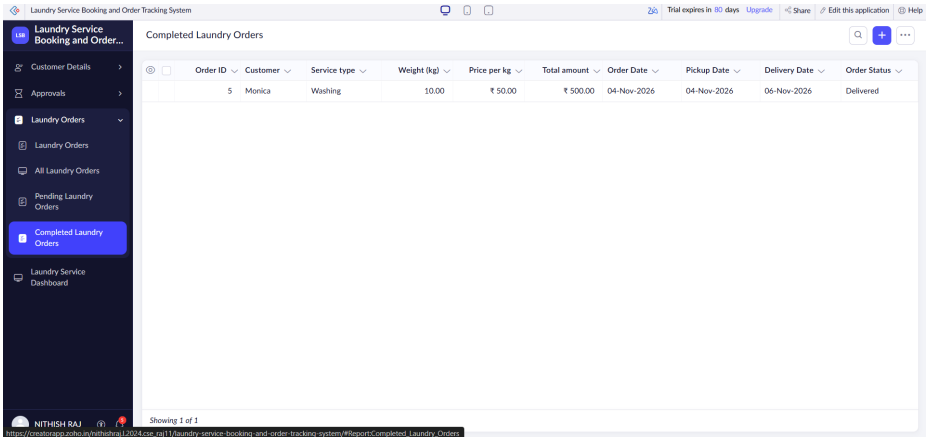


Figure 8.3: Step 3 — Delivered Order Appears in Completed Laundry Orders

Table 8.1: Sample Orders Saved in the Application

Order ID	Customer	Service type	Weight (kg)	Price per kg	Total amount	Status
1	Monica	Ironing	20	₹10	₹200	Received
2	NITHISH	Washing	10	₹50	₹500	Ready
3	Navin	Wash and Iron	20	₹60	₹1,200	Received
4	NITHISH	Dry Cleaning	10	₹60	₹600	Received
5	Monica	Washing	10	₹50	₹500	Delivered

Orders 1 to 4 were present when the dashboard in Figure 5.9 was captured (total ₹2,500); order 5 was delivered afterwards and appears in the Completed report.

8.2 Achievement of Objectives

Table 8.2: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
Structured data model and forms	Sections 4.2, 4.3 and 5.2	Two linked forms with ten field types; Figures 5.2–5.5
Calculation, discount and validation using Deluge	Sections 5.5 and 5.6	DS-01 to DS-05; Figures 5.10–5.14; test cases TC-03 to TC-07 passed
Order tracking with blueprint, approval and confirmation message	Section 5.6	Blueprint Received → Washing → Ready → Delivered and CEO approval; Figures 5.17–5.20 and 6.1
Reports and dashboard	Sections 5.3 and 5.4	Four list reports and a dashboard; Figures 5.6–5.9
Testing and governance	Chapters 6 and 7	10 of 10 test cases passed; security and governance documented

8.3 Comparison with the Existing System

Table 8.3: Existing System versus Proposed Application

Parameter	Existing System	Proposed Application	Improvement
Recording an order	Handwritten in a register	One form submission linked to a customer record	Structured, searchable record
Charge calculation	Worked out by hand	Total, discount and final amount filled by Deluge workflows	Fewer calculation errors
Order status	Verbal / memory	Blueprint stages with a status field	Status visible at any time
Summaries	Compiled manually	Pending and Completed reports and dashboard	Always based on current saved data
Data validation	None	Mandatory fields and weight/date validation	Invalid orders are rejected

8.4 Limitations of the Low-Code Approach

Zoho Creator's customisation options and the Deluge language do not offer the freedom of a fully custom application, and capabilities depend on the subscription edition. In this project the trial account supports a single user, so role-based access for staff, delivery agents and customers could not be demonstrated, and the three schedules were left disabled. Features such as payment tracking, customer feedback and delivery management would need additional forms or integrations, which are planned as future work.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

The forms, lookup, reports, dashboard and approval were created through no-code visual configuration. The calculations, discount rule, validations and confirmation message required low-code Deluge scripting. A payment gateway, an SMS service or a custom customer app would need pro-code development. The project shows that the simplest platform feature should be used where it meets the requirement and that code should be added only when it gives necessary functionality.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The Laundry Service Booking and Order Tracking System was developed using Zoho Creator to provide a simple and efficient way of managing a laundry business. It links customers and orders, calculates the total, discount and final amount automatically, validates weight and dates, tracks every order through Received, Washing, Ready and Delivered with a blueprint, supports CEO approval with an email to the submitter, and presents pending orders, completed orders and revenue through reports and a dashboard.

By using Zoho Creator's visual builders together with Deluge scripting, the application reduces manual work and repeated calculation with very little conventional programming. All ten functional test cases passed. The project demonstrates how low-code technology can solve a real business-management problem and provides a base for the enhancements below.

9.2 Future Enhancements

- Add a Service / Price List form so that the price per kg is filled automatically from the selected service.
- Add an Order Items subform to record individual garments and quantities.
- Add Pickup & Delivery, Payment Details and Customer Feedback forms with their reports.
- Implement role-based access for Staff, Delivery Agent and Customer (with a customer portal) as designed in Table 4.4.
- Enable the daily reminder, weekly report and monthly summary schedules, and add SMS, WhatsApp or email notifications to customers.
- Integrate an online payment gateway and add Zia-assisted analytics after validating their accuracy.

9.3 Learning Outcomes and Career Relevance

This project gave practical experience in converting a business problem into a data model, building forms, lookups, reports and a dashboard, writing Deluge scripts for calculation and validation, and configuring a blueprint and an approval. It also developed an understanding of testing, access control and deployment considerations. These skills are relevant to low-code developer, business-process automation, application support and solutions-consulting roles.

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APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created & sections/menu organised	Yes	Laundry Service Booking and Order Tracking System; 4 menu groups	5.1	Fig. 5.1
II	Forms with ≥ 8 distinct field types	Yes	Customer details and Laundry Orders (10 field types)	5.2	Figs. 5.2–5.5
II	Lookup field (relationship between forms)	Yes	Customer in Laundry Orders	4.2, 5.2	Fig. 5.4
II	Subform (parent–child data)	No	Not used in this version	—	—
II	Field validations / layout rules	Yes	Mandatory fields; 3 validation workflows	5.2, 5.5	Figs. 6.2, 6.3
II	List report with filter, grouping, custom action	Yes	Pending and Completed Laundry Orders (status filter)	5.3	Figs. 5.7, 5.8
II	Kanban / Calendar / Timeline / Map report	No	Not created	—	—
II	Pivot table / chart report	Yes	Orders by Status (pie), Revenue by Service Type (bar)	5.4	Fig. 5.9
II	Page / dashboard	Yes	Laundry Service Dashboard	5.4	Fig. 5.9
II	Deluge – form events (validate / success / field actions)	Yes	On user input, on validate and on success workflows	5.5	Figs. 5.10–5.14
II	Deluge – functions, fetch, loops, collections	Partial	Record fetch used in DS-05; no custom functions or loops	5.5	Fig. 5.14
II	Email / in-app notifications	Yes	Approval ‘Send mail’ actions	5.6, 5.7	Fig. 5.18
II	Form workflow	Yes	12 form workflows	5.6	Fig. 5.15
II	Scheduled workflow	Yes	3 schedules configured (currently disabled)	5.6	Fig. 5.16
II	Approval process	Yes	Laundry Order Approval (CEO)	5.6	Figs. 5.17, 5.18
II	Blueprint	Yes	Laundry Order Process	5.6	Figs. 5.19, 5.20, 6.1
II	Batch / report workflow	No	Not created	—	—
II	Testing & debugging evidence	Yes	10 test cases	6	Figs. 6.1–6.3
III	Integration with a Zoho service	Partial	Built-in email action only	5.7	—
III	Connection / REST API call (invokeurl)	No	Not used	—	—
III	Webhook or Creator REST API	No	Not used	—	—
III	Users, roles and permission profiles	Partial	Single user under the CEO role; other roles proposed	5.8	—

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
III	Record-level access criteria / customer portal	No	Not configured	—	—
III	Zia / AI feature	No	Not used	5.9	—
III	Analytics (charts)	Yes	Dashboard charts	5.9	Fig. 5.9
IV	Performance optimisation applied & measured	Partial	Design-level optimisations; not measured	7.2	—
IV	Deployment across environments	No	Single trial environment	7.1	—
IV	Security controls	Partial	Login, mandatory fields, validation, approval	7.3	—
IV	Governance (audit trail, administration)	Partial	Naming and ownership practices documented	7.4	—
V	Pro-code extension (Catalyst / widgets / custom functions)	No	Not built	5.10	—
V	Low-code limitations & spectrum analysis	Yes	Discussed	8.4, 8.5	—
—	Mobile app / portal access	Partial	Mobile/tablet layout views available; no portal	5.11	Figs. 5.1, 5.9

Total features fully implemented: 16 / 32; partially implemented: 7.

APPENDIX B – COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2	2 — Medium	
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6	3 — High	
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.7–5.9	1 — Low (email action only; single user)	
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7	2 — Medium	
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3	2 — Medium	

APPENDIX C – COMPLETE DELUGE SOURCE CODE

C.1 DS-01 – Calculate Total Amount on Weight Input (Laundry Orders › Created or Edited › User input of Weight (kg))

```
// Fetch the Price per kg from the current record context
pricePerKg = input.Price_per_kg;
// Fetch the Weight in kg from the current user input
weightKg = input.Weight_kg;
// Validate that both values are present before calculation
if(!pricePerKg.isNull() && !weightKg.isNull())
{
    // Calculate total amount by multiplying weight and price per kg
    totalAmount = weightKg * pricePerKg;
    // Set the calculated total amount to the Total_amount field
    input.Total_amount = totalAmount;
}
else
{
    // If either value is missing, set total amount to 0
    input.Total_amount = 0;
}
```

C.2 DS-02 – Apply Laundry Discount (Laundry Orders › Created or Edited › User input of Total amount)

```
// Calculate 10% discount if Total_amount >= 500, otherwise set discount to zero
totalAmount = input.Total_amount;
if(totalAmount >= 500)
{
    discountAmount = totalAmount * 0.10;
}
else
{
    discountAmount = 0;
}
// Set the Discounted_Amount field
input.Discounted_Amount = discountAmount;
// Calculate Final_Amount as Total_amount minus Discount_Amount
finalAmount = totalAmount - discountAmount;
// Set the Final_Amount field
input.Final_Amount = finalAmount;
```

C.3 DS-03 – Validate Laundry Weight (Validations on form submission)

```
if(input.Weight_kg <= 0)
{
    alert(Weight_kg,"Weight must be greater than zero");
    cancel submit;
}
```

C.4 DS-04 – Validate Pickup and Delivery Dates (Validations on form submission)

```
if(!isNull(input.Pickup_Date) && !isNull(input.Delivery_Date))
{
    if(input.Delivery_Date < input.Pickup_Date)
    {
        alert(Delivery_Date,"Delivery Date cannot be earlier than Pickup Date");
        cancel submit;
    }
}
```

C.5 DS-05 – Generate Order Confirmation Message (Successful form submission)

```
// Fetch customer details using the lookup ID
if(!input.Customer.isNull())
{
    customerRecord = Customer_details[ID == input.Customer];
    customerName = customerRecord.Customer_Name.first_name;
}
else
{
    customerName = "Unknown";
}
// Extract order details
orderID = input.Order_ID1;
orderDate = input.Order_Date;
serviceType = input.Service_type;
weight = input.Weight_kg;
finalAmount = input.Final_Amount;
pickupDate = input.Pickup_Date;
deliveryDate = input.Delivery_Date;
// Build the confirmation message
confirmationMessage = "Order Confirmation" + "\n" + "-------------------" + "\n"
    + "Order ID: " + orderID + "\n" + "Customer: " + customerName + "\n"
    + "Order Date: " + orderDate + "\n" + "Service Type: " + serviceType + "\n"
    + "Weight: " + weight + " kg" + "\n" + "Total Amount: " + finalAmount + "\n"
    + "Pickup Date: " + pickupDate + "\n" + "Delivery Date: " + deliveryDate + "\n"
    + "-------------------" + "\n" + "Thank you for your order!";
// Store the message in the Order_Message field
Order_Message = confirmationMessage;
```

Note: in C.5 the long concatenation expression is shown wrapped over several lines for readability; it is a single statement in the Deluge editor (Figure 5.14).

APPENDIX D – USER MANUAL

D.1 Accessing the Application

The Laundry Service Booking and Order Tracking System is developed using Zoho Creator and is used through a web browser.

- Open a web browser and sign in to the Zoho Creator account that owns the application.
- Open the application “Laundry Service Booking and Order Tracking System” and choose ‘Access this application’.
- The left menu shows Customer Details, Approvals, Laundry Orders and the Laundry Service Dashboard.

D.2 Instructions for the Owner / Administrator (CEO role)

The prototype is a single-user application, so the owner performs all of the following steps.

- **Step 1:** Open Customer Details › Customer details, enter the name, phone, email and address and click Submit.
- **Step 2:** Open Laundry Orders › Laundry Orders and select the customer and the service type.
- **Step 3:** Enter the weight and the price per kg. The total amount, discounted amount and final amount are filled automatically.
- **Step 4:** Enter the order, pickup and delivery dates and select the order status, then click Submit. If the weight is zero or the delivery date is before the pickup date, correct the entry and submit again.
- **Step 5:** To update progress, open the order record and click the blueprint button shown (Start Washing, Mark as Ready, Complete Delivery).
- **Step 6:** Open Approvals to approve or reject orders sent for review; an email is sent to the submitter.
- **Step 7:** Use All Laundry Orders, Pending Laundry Orders and Completed Laundry Orders to review orders.
- **Step 8:** Open the Laundry Service Dashboard to see total, pending and completed orders, revenue, orders by status and revenue by service type.

D.3 Instructions for Staff, Delivery Agent and Customer (proposed roles)

These roles are not configured in the prototype. In a multi-user version, Staff would book orders and update progress, the Delivery Agent would view assigned deliveries and update delivery status, and the Customer would place orders and view their own orders through a portal.

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	NITHISH RAJ L	Register No.	240701366
Project Title	Laundry Service Booking and Order Tracking System	Date	

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		**100**	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner
Name: _____

Signature of External Examiner
Name: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	NITHISH RAJ L	Register No.	240701366
Department / Section	Computer Science and Engineering	Date of Viva	For examiner to complete
Application URL Verified	Yes / No	Demo Duration	_____ min

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	**100**	

Examiners' Remarks:

_____

_____

INTERNAL EXAMINER
Name: _____

EXTERNAL EXAMINER
Name: _____